Loan & Financial Risk Analysis

Business Insights & Recommendations

Project	Credit risk analysis case (Nova Bank challenge), completed in Excel
Data	32,581 loan records, 29 columns, borrowers in the USA, UK and Canada
Type of analysis	Descriptive (data cleaning, exploratory analysis, dashboard). No predictive model was built.
Key sources	EDA documentation (Q1–Q7), 3-page Excel dashboard, handwritten project notes

1. Executive Summary

Nova Bank wants to lend fairly while protecting itself from unnecessary risk. This project looks at 32,581 loans to understand who tends to default and why that matters for lending decisions. The overall default rate is **21.8%** (7,108 defaulted and 25,473 non-defaulted loans), with an average loan of about $9,589 and an average borrower income of $66,075.

Objective. Identify borrower groups that are more or less likely to default, explore how loan size, income and debt burden relate to repayment, spot early warning signs, and suggest how lending policy could be adjusted.

Main results. The strongest differences in default rates appear for loan grade (9.96% for A to 98.44% for G), debt-to-income band (11.07% to 56.65%), previous default on file (37.81% vs 18.39%), home ownership (7.47% for owners to 31.57% for renters) and employment length (27.94% for new employees to 16.23% for 11+ years). Age, education, marital status, country, loan term and employment type show only small differences.

Important note. These are patterns in historical data. They show association, not cause, and they are not predictions of future defaults.

2. Business Questions

#	Question	Why it matters for credit risk
1	Which types of borrowers are more likely to default?	Shows where losses concentrate and which profile details are worth collecting.
2	Do certain loan purposes carry more risk?	Helps decide where to tighten checks or adjust pricing.
3	How do loan-to-income and debt-to-income relate to repayment?	Tests the basic idea of repayment capacity.
4	How do past defaults or credit history length affect outcomes?	Tests whether past behaviour signals future behaviour.
5	Are there differences between the USA, UK and Canada?	Shows if one set of standards can work in all markets.
6	Which loan grades or terms are safer or riskier?	Checks whether existing grades separate risk.
7	Does employment type or home ownership matter?	Looks for stability indicators that are easy to collect.
8	Can “safe” and “risky” groups be identified?	Brings the findings together into practical segments.

Data preparation in brief. Invalid age values (such as 123 and 144) and invalid employment lengths (123 years) were replaced with the median. 895 missing employment lengths were also replaced with the median; 4,105 zero values were kept as real values (new employees). 3,116 missing interest rates were filled with the median rate of the same loan grade. The original loan-to-income column was inconsistent with income, so it was recalculated as loan amount divided by income. Text columns such as loan intent and employment type were standardised, and city, latitude and longitude were removed.

3. Key Findings & Insights

Findings are ordered by strength of the difference in default rates. Each separates what was observed from interpretation and limitations.

Finding 1: Loan grade separates risk very strongly

Analytical question: Which loan grades or terms seem safer, and which are riskier?

Observed: Default rate is 9.96% for grade A, 16.28% for B, 20.73% for C, then 59.05% for D, 64.42% for E, 70.54% for F and 98.44% for G. By term, default rates range only from 19.94% (12 months) to 22.34% (60 months), with no consistent pattern.

Interpretation: Grades D to G are very risky compared with A to C, and loan term adds little information on its own.

Business significance: The existing grading system appears to rank risk well in this portfolio. Grades D to G deserve stricter review and closer monitoring.

Limitations: Grade is probably assigned using other borrower information, so it overlaps with other variables and should not be read as an independent cause. The number of loans in each grade is not available in the documentation, so the share of the portfolio in D to G is unknown.

Finding 2: Debt burden is higher among borrowers who defaulted

Analytical question: How do loan-to-income (LTI) and debt-to-income (DTI) relate to repayment?

Observed: Defaulted borrowers had an average DTI of 42% vs 32% for non-defaulted, and an average LTI of 25% vs 15%. On the dashboard, default rate by DTI band is 11.07% (low, under 20%), 13.86% (moderate, 20–34%), 23.73% (elevated, 35–50%) and 56.65% (high, over 50%).

Interpretation: The more of a borrower's income is already committed to debt or to the new loan, the higher the observed default rate. The rise is steepest above 50% DTI.

Business significance: DTI and LTI are available at application time and map directly onto affordability checks.

Limitations: The DTI band values were read from a pie chart whose legend order was matched to the slices, so please verify them in the workbook. Averages hide the spread inside each group, and DTI and income may be related to each other.

Finding 3: Previous default on file is a clear warning sign

Analytical question: How do past defaults or longer credit histories affect loan outcomes?

Observed: Borrowers with a previous default have a 37.81% default rate vs 18.39% for those without. Within credit-history categories, the range is 35.82% to 38.48% with a previous default and 16.77% to 19.15% without. By credit history length, default rates are close: long 20.65%, moderate 20.64%, short 22.52%, very long 21.72%.

Interpretation: A previous default roughly doubles the observed default rate, whatever the credit history length. Length of history alone shows no steady pattern.

Business significance: A previous default flag is cheap to check and could trigger a deeper review. Credit history length should not be used alone to judge a borrower.

Limitations: Past delinquencies were discussed in the notes as showing "somewhat higher" default rates, but no figures were provided, so no number is quoted here.

Finding 4: Renters make up half the loans and most of the defaults

Analytical question: Does home ownership make a difference?

Observed: Default rate is 7.47% for owners, 12.57% for mortgage holders and 31.57% for renters (16,446 loans). The “Other” group has 30.84% but only 107 loans. A rough calculation (16,446 x 31.57%) gives about 5,190 defaults for renters, around 73% of all 7,108 defaults. This is my own approximation from the reported figures.

Interpretation: Renters are both the largest group and the highest-default group among the main categories.

Business significance: Home ownership is a simple stability indicator. The reason renters default more should be studied before it is used in decisions.

Limitations: Home ownership may stand in for wealth, income or age. It should not replace affordability checks, and banks should confirm that using it is allowed and fair in each market.

Finding 5: Short employment length goes with higher default rates

Analytical question: Does employment matter?

Observed: Default rate by employment length: 0 years 27.94%, 1–2 years 26.51%, 3–5 years 21.01%, 6–10 years 18.07%, 11+ years 16.23%. Employment type, in contrast, shows little difference (21.57% full-time to 22.67% unemployed).

Interpretation: Time in current job matters more than the type of employment, and the rate falls steadily with longer service.

Business significance: Job stability could be an extra input for newer employees, for example by asking for more proof of income stability.

Limitations: 895 missing employment lengths were replaced with the median, which may slightly flatten differences. The dashboard labels “5-3” and the 0-year group (4,105 records) should be double-checked. This finding is not discussed in the written EDA answers.

Finding 6: Loan purpose: debt consolidation, medical and home improvement carry the most risk

Analytical question: Do certain loan purposes carry more risk?

Observed: Default rate: debt consolidation 28.59% (5,212 borrowers), medical 26.70% (6,071), home improvement 26.10% (3,605), personal 19.89%, education 17.22%, venture 14.81% (5,719). Medical loans produce the largest number of defaults, about 1,621 (6,071 x 26.70%).

Interpretation: Debt consolidation is riskiest per loan. Medical is the biggest contributor to total defaults because of volume.

Business significance: Review criteria and monitoring for these three purposes. A low rate for venture and education does not justify looser standards.

Limitations: Purpose may reflect borrower financial stress rather than cause default. Personal and education loan counts are not given in the notes.

Finding 7: Income band shows a large gap (needs verification)

Analytical question: Which types of borrowers are more likely to default?

Observed: The documentation reports an 81.03% default rate for low-income borrowers and 8.78% for affluent borrowers. These figures were garbled in the PDF and cannot be cross-checked on the dashboard.

Interpretation: If correct, income is one of the strongest borrower-level indicators, but grade, DTI and home ownership also show wide gaps, so income should not be called the single most important factor.

Business significance: Income should be considered together with debt and loan size.

Limitations: Confirm both values and the band definitions before using them. Income also moves together with DTI and LTI, because both ratios use income.

Finding 8: Several variables show little difference

Analytical question: Are there clear differences by demographics or country?

Observed: Age: 20–24 is highest at 23.22% (45+ is 22.65%, lowest is 20.05% for ages 30–34). Education: 21.16% to 22.17%. Marital status: 21.37% to 21.96%. Country: USA and Canada 21.86%, UK 21.73%. Average income, loan amount, DTI and LTI are also very similar across the three countries.

Interpretation: In this dataset these variables add little to explaining default. Differences of one or two percentage points are close to flat.

Business significance: Credit decisions can focus on financial and credit variables. Using age or marital status as drivers would add little and could raise fairness concerns.

Limitations: Small differences could still be meaningful in a larger sample, and the analysis did not test statistical significance.

4. Borrower Risk Profile

Answer to Question 8: Yes. Even with a one-variable-at-a-time analysis, the data suggests groups of borrowers that look clearly lower or higher risk. The table compares them. Each row is a separate view of the same loans, not a separate group of people.

Indicator	Lower-risk group (default rate)	Higher-risk group (default rate)	Gap (pts)
Loan grade	A (9.96%), B (16.28%)	D to G (59.05% to 98.44%)	88.5 (A vs G)
Debt-to-income	Under 20% (11.07%)	Over 50% (56.65%)	45.6
Previous default	None (18.39%)	Yes (37.81%)	19.4
Home ownership	Own (7.47%), mortgage (12.57%)	Rent (31.57%)	24.1
Employment length	11+ years (16.23%)	0 years (27.94%), 1–2 years (26.51%)	11.7
Loan purpose	Venture (14.81%), education (17.22%)	Debt consolidation (28.59%)	13.8
Income (to verify)	Affluent (8.78%)	Low income (81.03%)	72.3

Little or no separation: marital status (0.6 pts), country (0.1 pts), education (1.0 pt), employment type (1.1 pts), credit history length (1.9 pts), loan term (2.4 pts) and age (3.2 pts).

A practical "safe" profile (based on these results) would combine a grade of A or B, DTI below 20%, no previous default, homeownership and long employment. **A "risky" profile** would combine grade D to G, DTI over 50%, a previous default, renting and under three years of employment.

Caution: the project did not combine these factors (for example, default rate for renters with DTI over 50%). The profiles are an informed summary of single-variable results, not tested segments. Grade likely overlaps with the other indicators, so the gaps cannot simply be added together.

5. Financial Risk Analysis

Variable	What the analysis shows	Status
Debt-to-income (DTI)	Average 42% (defaulted) vs 32%; default rate rises from 11.07% to 56.65% across bands. Portfolio average 35%.	Supported
Loan-to-income (LTI)	Average 25% (defaulted) vs 15%. Recalculated from loan amount divided by income after the original column proved inconsistent.	Supported
Interest rates	Portfolio average is 11.0%. No default rate by interest rate is given. 3,116 missing rates were filled using the grade median, so any rate and grade link would be partly built in.	Not analysed
Loan amounts	Average is about $9,589. By country: USA $9,553 (lowest), Canada $9,642 (highest). Charts of average loan by DTI band and income band have no data labels. No link to default is quantified.	Not analysed
Borrower income	Average $66,075 (UK highest at $66,306). Income-band default rates reported as 81.03% (low) and 8.78% (affluent).	Needs verification
Credit history	Length shows weak differences (20.64% to 22.52%). Previous default is strong (37.81% vs 18.39%).	Supported
Past delinquencies	Described as linked to “somewhat higher” default rates, with no figures.	Limited
Loan grade	9.96% (A) to 98.44% (G).	Supported
Loan intention	Debt consolidation 28.59% highest; venture 14.81% lowest.	Supported

6. Business Recommendations

Recommendation 1: Apply tiered credit review by loan grade

Supporting insight	Grades D to G default at 59.05% to 98.44%, vs 9.96% for A.
Business rationale	Concentrating review effort where risk is highest uses staff time well.
How to apply	Require enhanced underwriting, senior approval or lower exposure limits for D to G applications. Keep a standard process for A to C.
KPI to monitor	Default rate by grade each quarter; share of new loans in D to G.

Recommendation 2: Set affordability checks using DTI and LTI

Supporting insight	Defaulted borrowers had DTI of 42% vs 32% and LTI of 25% vs 15%; the default rate is 56.65% for DTI over 50%.
Business rationale	Affordability is a direct measure of repayment capacity.
How to apply	Add review triggers (for example, DTI above 35% for extra checks and above 50% for manual approval). Test cut-offs on the bank’s own data before use.
KPI to monitor	Share of portfolio with DTI over 50%; default rate in each DTI band.

Recommendation 3: Flag previous default for mandatory review

Supporting insight	Previous default: 37.81% vs 18.39%, in every credit-history category.
Business rationale	It is a simple and reliable flag, and credit history length is not.
How to apply	Add the flag to the application workflow; request explanation and extra evidence of repayment. Do not use history length alone.
KPI to monitor	Default rate of flagged vs non-flagged loans; approval rate of flagged applicants.

Recommendation 4: Strengthen assessment for higher-risk loan purposes

Supporting insight	Debt consolidation 28.59%, medical 26.70% (largest default volume, about 1,621), home improvement 26.10%.
Business rationale	These purposes combine high rates or large volumes.
How to apply	Review criteria, loan-size limits and pricing for these purposes. Monitor medical loans by volume.
KPI to monitor	Default rate and default count by purpose.

Recommendation 5: Use home ownership and employment length as supporting context

Supporting insight	Renters 31.57% vs owners 7.47%; 0–2 years employment 26.5% to 27.9% vs 16.23% for 11+ years.
Business rationale	They are easy to collect and show clear differences, but may reflect other factors.
How to apply	Ask renters and new employees for more proof of stable income. Check legal and fairness rules before any use in scoring.
KPI to monitor	Default rate for renters and 0–2 year employees; approval-rate gaps across groups.

Recommendation 6: Do not rely on age, marital status, education, country or term

Supporting insight	These show differences of about 0.1 to 3 points.
Business rationale	They add little predictive information and may raise fairness concerns.
How to apply	Keep one core assessment standard across the USA, UK and Canada, and monitor markets separately.
KPI to monitor	Default rate by country each quarter; consistency of approval criteria.

These recommendations are based on associations in one dataset. Thresholds and policies should be tested on the bank's own data before being adopted.

7. Limitations & Further Analysis

Limitations of the dataset and analysis

- Single-variable views: the project did not test combinations of variables, so overlaps (for example grade and DTI) are not controlled.
- Cleaning choices matter: median replacements (ages, employment length) and grade-median interest rates could reduce natural variation.
- No significance testing, and some groups are small (the "Other" home ownership group has 107 loans).
- No data on interest rate, loan amount or income compared with default status; no loss amounts or recovery information.
- Income-band figures and the DTI band values on the dashboard need verification.
- The data is a snapshot with no time dimension, so trends and changes in default behaviour cannot be seen.

What cannot be concluded

- That any factor causes default, or that changing it would reduce default.

- That the patterns would hold for new applicants or other banks.
- That a borrower with a given profile will default.

Additional analysis that could improve the project

- Cross-tabulate key variables (grade by DTI, home ownership by income) to test the “safe” and “risky” profiles.
- Compare average interest rate, loan amount and income for defaulted vs non-defaulted loans.
- Report portfolio share and default counts by grade, and total loan amount at risk.
- Fix dashboard labels and replace the pie charts of default rates with bar charts.

Possible future predictive modelling (not done in this project)

A logistic regression or decision-tree model could estimate default probability and test which variables stay important when others are controlled. This would need a train/test split, a check for overlap between grade and other variables, and fairness review. None of this has been done here.

8. Final Conclusion

The analysis shows that default risk in this portfolio is not spread evenly. Loan grade, debt burden, previous default, home ownership, employment length and loan purpose all show meaningful differences, while age, education, marital status, country and loan term show little. For a bank, this means credit assessment can focus on repayment capacity and past credit behaviour rather than on demographic or geographic features.

The value of the work is in turning raw data into clear questions, checked numbers and practical, measurable actions, while being open about what the data cannot prove. The next step is to test combined risk factors and, later, to build a predictive model.